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FIRST SEMESTER B.Tech EXAMINATION, JANUARY-2025

MATRIX ALGEBRA AND CALCULUS

Time: 3 Hours **Maximum Marks: 100**

Instructions: (i) Missing data may be suitably assumed

(ii) Answer all Questions

		Questions	Blooms Level	CO's	Marks
		PART-A	**5 x2=10**		
1.	**a.**	Calculate the rank of matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$	L2	CO1	**02**
	b.	Find the general solution of the differential equation $4y'' + 4y' - 3y = 0.$	L2	CO1	**02**
	c.	Calculate the first and second-order partial derivatives of the function $z = x^3 + y^3 - 3axy$.	L2	CO1	**02**
	d.	If $f(x,y,z) = 2x + yz - 3y^2$, then find $grad\ f$.	L2	CO1	**02**
	e.	Solve $\int_0^{\pi/2} \sin^6 x\, dx$.	L2	CO1	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Determine the rank of matrix $A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$ by reducing it to row Echelon form.	L2	CO2	**05**
		OR			
	b.	Test the consistency and solve the following system of equations $x + y + z = 6; x - y + 2z = 5;\ 3x + y + z = 8$.	L3	CO2	**05**
3.	**a.**	Solve $D^3 - 2D^2 - 5D + 6 = e^{4x} + 1$	L3	CO3	**05**
		OR			
	b.	Solve using the method of variation of parameters $y'' + 4y = \tan 2x.$	L3	CO3	**05**
4.	**a.**	If $u = x + 3y^2 - z^3, v = 4x^2yz, w = 2z^2 - xy$, then find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$, at (1,-1,0)	L3	CO4	**05**
		OR			
	b.	Prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$, where $u = (x-y)^4 + (y-z)^4 + (z-x)^4$.	L2	CO4	**05**

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5.	a.	Find the components of velocity and acceleration at $t = 2$ on the curve $\vec{r} = (t^2+1)\hat{\imath} + (4t-3)\hat{\jmath} + (2t^2-6t)\hat{k}$ in the direction of the vector $\hat{\imath} + 2\hat{\jmath} + 2\hat{k}$.	L3	CO5	05
		OR			
	b.	Calculate directional derivatives of $\varphi = 4xz^3 - 3x^2y^2z$. At (2,-1,2) along $2\hat{\imath} - 3\hat{\jmath} + 6\hat{k}$.	L2	CO5	05
6.	a.	Solve the integral $\int_0^{\pi} x \sin^7 x \, dx$ using Reduction formula.	L2	CO6	05
		OR			
	b.	Solve the integral $\int_0^{\infty} \frac{dx}{1+x^4}$ using Beta-Gamma function.	L3	CO6	05
7.	a.	Solve $(D^2 - 4D + 1)y = e^{2x} \sin 2x$.	L3	CO3	05
		OR			
	b.	Write MATLAB code to find the first and the second-order partial derivatives of the function $f(x,y) = x^3 + 3xy - 15x^2 - 15y^2 + 72x - 4y$.	L2	CO1	05
		PART-C	**6 X 10=60**		
8.	a.	Determine the Eigenvalues and the corresponding Eigenvectors of the matrix $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$.	L3	CO2	10
		OR			
	b.	Employ Gauss–Seidel method to solve the following system of linear equations. $5x+2y+z=12, x+4y+2z=15, and\ x+2y+5z=20$ Carry out four iterations taking the initial approximation to the solution as (1,0,3).	L3	CO2	10
9.	a.	Solve $X' = \begin{bmatrix} -5 & 1 \\ 4 & -2 \end{bmatrix} X$ by diagonalization and hence discuss stability.	L3	CO3	10
		OR			
	b.	Solve $(1+x)^2 \frac{d^2y}{dx^2} + (1+x)\frac{dy}{dx} + y = 2\sin[\log(1+x)]$.	L3	CO3	10
10.	a.	Prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{(x+y+z)}$, where $u = log(x^3 + y^3 + z^3 - 3xyz)$ and hence show that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = \frac{-9}{(x+y+z)^2}$.	L3	CO4	10
		OR			
	b.	Investigate the extreme values of the given function $sinx\ siny\ sin(x+y)$, where $0 < x < \frac{\pi}{2}, 0 < y < \frac{\pi}{2}$.	L3	CO4	10

11.	**a.**	Show that the vector point function $\vec{v} = \frac{x\hat{i}+y\hat{j}}{x^2+y^2}$ is both solenoidal and irrotational.	L3	CO5	**10**
		OR			
	b.	Show that the vector point function $\vec{F} = (2xy^2z)\hat{i} + (x^2z^2 + z\cos(yz))\hat{j} + (2x^2yz + y\cos(yz))\hat{k}$ is a conservative force field? Find its scalar potential φ such that $\nabla\varphi = \vec{F}$.	L3	CO5	**10**
12.	**a.**	Solve the integral i. $\int_0^{\pi} \frac{\sin^4\theta}{(1+\cos\theta)^2} d\theta$ ii. $\int_0^{2a} x^2\sqrt{2ax - x^2}dx$	L2	CO6	**10**
		OR			
	b.	Show that $\int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{\sin t}} dt \times \int_0^{\frac{\pi}{2}} \sqrt{\sin t}\, dt = \pi$	L3	CO6	**10**
13.	**a.**	The temperature T at any point (x, y, z) in space is given by $T = 400xyz^2$. Find the highest temperature at the surface of a unit sphere $x^2 + y^2 + z^2 = 1$.	L3	CO4	**10**
		OR			
	b.	Write MATLAB code to create the matrix $A = \begin{bmatrix} 1 & 5 & 8 \\ 0 & 4 & 3 \\ 2 & 3 & 6 \end{bmatrix}$ and perform the following operations: **i.** Determinant of A **ii.** Rank of A **iii.** Size of matrix A **iv.** Characteristic polynomial of the square matrix A **v.** Eigenvalues of A	L2	CO1	**10**

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